

Uncertainty-Aware Airlight-Aligned Dual-Gain Recovery with Exact RGB Feasibility for Training-Free Single-Image Dehazing

Maksim Sitnikov

Preprint draft, August 2026

Abstract

The inverse atmospheric-scattering model amplifies transmission, airlight, and sensor-noise errors when a scalar gain $1/t$ is applied in dense haze. We introduce A²CR, a training-free recovery operator that decomposes the airlight-centered RGB residual into components parallel and perpendicular to the airlight vector. Two gains are obtained from analytic quadratic risks containing transmission, airlight, and noise uncertainty. Each gain pair is constrained by the exact convex polygon induced by the six RGB-cube inequalities and gain bounds. An edge-aware joint total-variation problem is solved with a primal-dual algorithm and exact polygon projection. We additionally study C³R-HSV, a complementary recovery in continuous $(V, SV \cos H, SV \sin H)$ coordinates with three gains and a non-empty second-order-cone corridor. On a controlled DIODE RGB-D evaluation comprising 500 frames and 22,500 reproducible haze recipes, the full variant improves over scalar inversion by 2.951 dB PSNR (95% bootstrap CI [2.900, 3.001]), 0.191 SSIM [0.187, 0.196], and -0.978 CIEDE2000 $[-1.074, -0.883]$, while eliminating post-recovery RGB violations. Results on four real paired datasets are mixed and were not blind: the method is strongest on Dense-Haze but does not dominate RFEP, DCP, and Haze-Lines elsewhere. A frozen O-HAZE pilot further shows that cylindrical feasibility eliminates clipping but currently under-recovers compared with the RGB method; its uncertainty penalties are not supported by that pilot. We therefore claim an interpretable constrained-recovery study with an explicit negative branch, not state-of-the-art dehazing. Code, tests, manifests, and statistical artifacts accompany the paper.

1 Introduction

Single-image dehazing is ill-posed: the observed linear-radiance color

$$I(x) = J(x)t(x) + A(x)(1 - t(x)) \quad (1)$$

does not identify scene radiance J , transmission t , and airlight A from one RGB sample. Classical priors such as the dark channel prior (DCP) [1], boundary constraints [2], color attenuation [3], and haze-lines [4] make this ambiguity tractable without training. However, after any

estimate $\hat{t}, \hat{A}$ has been obtained, the common inverse $J = \hat{A} + (I - \hat{A})/\max(\hat{t}, t_{\min})$ applies the same gain to all color directions. In dense haze, that gain amplifies both model error and chromatic sensor noise.

We study the recovery operator itself. The key observation is geometric: errors parallel to the airlight vector and errors that change chromatic direction need not have the same reliability. The proposed method applies separate analytic gains, but never permits them to leave the RGB cube. Our contributions are:

- an airlight-aligned two-component recovery with gains derived from explicit quadratic uncertainty risks;
- an exact per-pixel convex feasible polygon for the two gains, rather than post-hoc RGB clipping;
- a joint edge-aware TV formulation solved by primal-dual splitting with exact polygon projection;
- a complementary circular-HSV recovery and guaranteed non-empty saturation/value cone, evaluated as a negative ablation rather than presented as an additional winner;
- controlled and real-paired evaluations that report uncertainty, runtime, negative results, and the non-blind status of development datasets.

We also report CAR-Dehaze, a separate case study for a weak red wall in O-HAZE scene 08. It motivates local color objectives but is not used as evidence of generalization.

2 Related Work

Classical priors. DCP estimates haze thickness from dark pixels in local haze-free patches [1]. Meng et al. constrain transmission through RGB boundaries and contextual regularization [2]. CAP estimates depth from HSV value and saturation [3], whereas non-local dehazing clusters clear colors into haze-lines in RGB space [4]. Bounded channel differences provide another deterministic color constraint [5].

Spatial airlight and color. Airlight-field estimation relaxes the constant-airlight assumption [6]. The color-constrained dehazing model jointly regularizes local airlight

and transmission and uses a learned natural-color distribution [7]. These works motivate spatial models, but neither formulates two airlight-aligned recovery gains with an exact RGB-feasible set.

Uncertainty and learned dehazing. Semi-UFormer learns pixel uncertainty in a teacher-student model [8], while BPUL uses learned perturbation-induced uncertainty and degradation consistency [9]. Recent methods emphasize domain adaptation, diffusion, or unpaired transport [10–12]. Our goal differs: no training, deterministic CPU execution, and an uncertainty model exposed in equations and diagnostic maps.

Circular HSV representations. RSVT translates points in the saturation-value plane while largely preserving hue [13]. HVI polarizes hue and saturation to remove the red seam and suppress unstable dark colors for low-light enhancement [14]. CIM-D uses a Cartesian sine/cosine HSV embedding and a physics-guided HSV decomposition network for dehazing [15]. Consequently, circular HSV embedding itself is not our novelty claim. We study whether analytic uncertainty gains and an explicit feasible cone add value in a deterministic, training-free recovery; the initial experiment below is deliberately negative.

A targeted search of arXiv, CVF Open Access, and author publication pages through 30 July 2026 found no direct analogue combining all five elements above. This is a bounded literature-search statement, not proof of worldwide priority.

3 Method

All physical operations use linear RGB. The final result alone is encoded to sRGB.

3.1 Airlight-aligned decomposition

For a global airlight estimate $A \in \mathbb{R}^3$, let $u = A/\|A\|_2$, $d = I - A$, and

$$p = uu^\top d, \quad q = (I_3 - uu^\top)d. \quad (2)$$

We recover

$$J = A + g_\parallel p + g_\perp q. \quad (3)$$

The parallel component primarily changes distance from the airlight; the perpendicular component changes chromatic direction.

3.2 Analytic uncertainty risk

For either component, with local signal energy S , mean transmission $\bar{t}$, transmission variance σ_t^2 , effective noise energy N , and identity prior U , we minimize

$$R(g) = S[(g\bar{t} - 1)^2 + g^2\sigma_t^2] + Ng^2 + U(1 - g)^2. \quad (4)$$

The unconstrained minimizer is

$$g^* = \frac{S\bar{t} + U}{S(\bar{t}^2 + \sigma_t^2) + N + U}, \quad 1 \leq g \leq 1/t_{\min}. \quad (5)$$

Independent risks use component energies and projected airlight/noise uncertainty. When uncertainty and noise vanish, Eq. (5) reduces exactly to $1/t$; this limit is unit-tested.

Transmission uncertainty is estimated from an ensemble of DCP bootstrap, CAP, and a simplified haze-line map. A weighted median and median absolute deviation are computed in optical depth $D = -\log t$, then propagated as $\sigma_t^2 \approx t^2\sigma_D^2$. Airlight uncertainty is the component-wise variance of patch/top-percent bootstrap estimates.

3.3 Exact RGB-feasible gain polygon

For each pixel and channel c , Eq. (3) must satisfy

$$0 \leq A_c + g_\parallel p_c + g_\perp q_c \leq 1. \quad (6)$$

Together with $1 \leq g_\parallel, g_\perp \leq g_{\max}$, the six half-planes in Eq. (6) form a non-empty convex polygon because (1, 1) reproduces the valid input. We enumerate feasible boundary intersections and edge projections in two dimensions and select the Euclidean projection of the proposed gain pair. Randomized tests cover 20,000 direct projections and 5,000 cached-polygon comparisons. No projected output violated the RGB cube.

3.4 Joint edge-aware TV

Let K_x be the feasible polygon at pixel x . We solve

$$\begin{aligned} \min_{g_\parallel, g_\perp} \quad & \sum_x R_{\parallel, x}(g_{\parallel, x}) + R_{\perp, x}(g_{\perp, x}) + \mu(g_{\parallel, x} - g_{\perp, x})^2 \\ & + \lambda \sum_x w_x (\|\nabla g_{\parallel, x}\|_2 + \|\nabla g_{\perp, x}\|_2), \\ \text{s.t.} \quad & (g_{\parallel, x}, g_{\perp, x}) \in K_x \quad \forall x. \end{aligned} \quad (7)$$

where w_x decreases across input-luminance edges. A Condat–Vu-style primal-dual splitting algorithm [16] handles the smooth quadratic risk explicitly, TV through dual-ball projection, and feasibility through the exact K_x projection. Step sizes use a local Hessian Lipschitz bound and $\|\nabla\|^2 \leq 8$. The best feasible iterate by the original objective is returned. A numerical unit test verifies objective decrease and feasibility at every pixel.

3.5 Complementary cylindrical recovery

To test whether the same risk-and-feasibility principle transfers to a color-oriented geometry, C³R-HSVmaps linear-RGB-derived HSV to

$$z = (V, c_x, c_y), \quad (c_x, c_y) = SV(\cos H, \sin H). \quad (8)$$

This removes the 0/360° seam and collapses hue as $SV \rightarrow 0$. Relative to the airlight point, chroma is decomposed into

radial and tangential components, producing three risk gains g_V, g_r, g_τ using Eq. (5). CAP remains evaluated in its calibrated sRGB/HSV space, whereas DCP, airlight, and recovery use linear radiance. CAP/DCP disagreement is measured as the equal-prior optical-depth variance; it is kept independent of the mean-fusion weight so that an endpoint weight cannot erase uncertainty.

The candidate is restricted to

$$K_x = \{(V, c) : 0 \leq V \leq 1, \|c\|_2 \leq S_{\max, x} V\}. \quad (9)$$

Because $S_{\max, x} \geq S_{\text{input}, x}$, the input is always in K_x . We find the largest $\alpha \in [0, 1]$ for which the segment from input to candidate remains in the cone, then apply a circular hue trust region without changing chroma magnitude. This guarantees representable HSV output, but it is not an exact reparameterization of Eq. (1): HSV is nonlinear and the three scalar gains are an approximate color recovery except in restricted cases such as neutral airlight within a fixed hue sector. A randomized 20,000-case test verifies cone feasibility and segment preservation.

4 Experimental Protocol

Controlled RGB-D synthesis. DIODE [17] supplies RGB, depth, and validity masks. We deterministically select 250 indoor and 250 outdoor validation frames. Whole physical scene groups are assigned to development/validation/internal-test splits (295/104/101 frames) without scene leakage. For each frame we synthesize five target transmissions, three airlights, and three noise recipes in linear RGB, yielding $500 \times 45 = 22,500$ recipes. Stable SHA-256-derived seeds make every streamed sample reproducible without storing generated images.

We compare scalar inversion (B0), scalar RGB boundary projection (B3), dual gain with uncertainty (B7), exact feasibility (B8), and joint TV (B9). Metrics include PSNR, SSIM [18], CIEDE2000 [19], chroma/hue error, RGB violations, flat-region noise residual, and runtime. A 10-frame, 4,500-row pilot additionally computes LPIPS AlexNet v0.1 [20].

Uncertainty. The 45 recipes are first averaged within each frame. We then run 10,000 indoor/outdoor-stratified paired bootstrap repetitions over 500 frames. Because DIODE validation contains only six physical scene groups, we additionally report leave-one-scene-group-out ranges. Real-paired comparisons use paired bootstrap by image identifier.

Real paired data. We use fixed test splits from I-HAZE [21], O-HAZE [22], Dense-Haze [23], and NH-HAZE [24]. These datasets were inspected during development; their results must not be called blind tests.

Table 1: DIODE controlled means over 22,500 recipes.

	PSNR↑	SSIM↑	DE00↓	Chroma↓	Invalid↓	ms↓
B0 scalar	18.797	0.515	12.521	12.672	0.259	0.30
B3 boundary	19.965	0.611	13.758	16.556	0	0.35
B7 uncertain	19.257	0.514	11.303	8.968	0.227	0.73
B8 feasible	20.723	0.602	11.951	11.894	0	1.01
B9 joint TV	21.748	0.707	11.543	11.372	0	31.18

Table 2: A²CR real-paired fixed test splits (not blind).

Dataset	n	PSNR↑	SSIM↑	DE00↓	LPIPS↓
I-HAZE	15	16.616	0.822	13.659	0.280
O-HAZE	22	17.046	0.798	14.342	0.331
Dense-Haze	27	12.480	0.478	22.110	0.692
NH-HAZE	27	13.228	0.602	20.901	0.462

Cylindrical pilot protocol. We use the fixed odd/even O-HAZE validation/test split at maximum dimension 192. Five C³R configurations are compared on 23 validation images. One configuration is selected by aggregate PSNR/SSIM/CIEDE2000 and evaluated once on the 22 test images. A²CR, HSV²CR, and default C³R use fixed defaults on the same test split. This small non-blind pilot is diagnostic and is not pooled with the native-resolution results in Table 2.

5 Results

5.1 Controlled ablation

Table 1 summarizes 112,500 valid rows. B8 and B9 eliminate all RGB violations. B9 has the best PSNR, SSIM, and flat-noise residual; B7 retains the best mean CIEDE2000 and chroma error, so no universal color claim is supported.

Against B0, B9 improves PSNR in 22,201/22,500 recipes, SSIM in all recipes, CIEDE2000 in 18,439, and chroma error in 17,082. Figure 1 shows frame-level intervals. All displayed quality intervals exclude zero; the approximately 31-fold cost over B8 is the main trade-off.

The LPIPS pilot produced 4,500/4,500 finite values. B9 obtained PSNR 21.264, SSIM 0.595, CIEDE2000 13.129, and LPIPS 0.624, compared with B0’s 18.281, 0.358, 14.359, and 0.865.

5.2 Real paired regression evaluation

Table 2 reports fixed defaults. A²CR is best among the four tested methods on Dense-Haze, but not elsewhere. On I-HAZE its LPIPS is significantly worse than RFEP in the paired bootstrap; on O-HAZE SSIM improves while LPIPS worsens. NH-HAZE improvements depend on comparator and metric. These negative results limit the claim to the recovery operator and controlled failure modes.

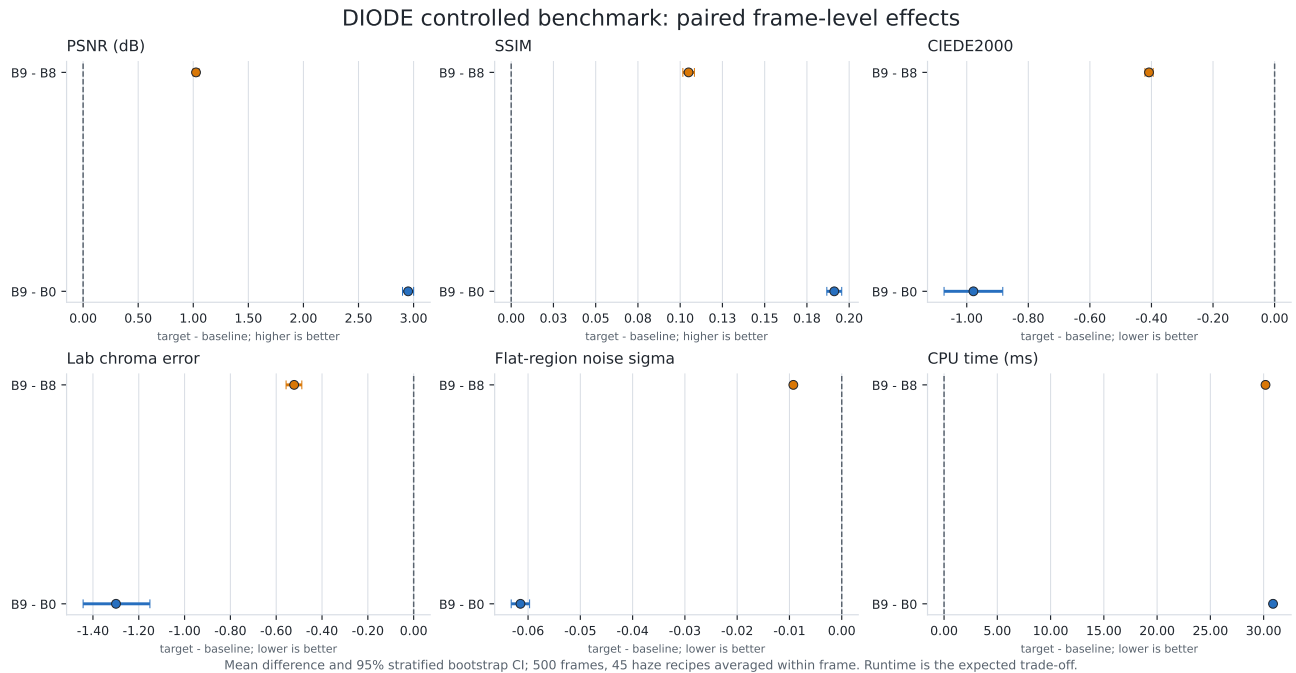


Figure 1: Paired frame-level effects with 95% stratified bootstrap intervals. Forty-five haze recipes are averaged within each of 500 frames before resampling. Lower is better for CIEDE2000, chroma error, noise, and runtime.

Table 3: O-HAZE test pilot at maximum dimension 192. C³R-v2 was selected only on validation.

Method	PSNR $\uparrow$	SSIM $\uparrow$	DE00 $\downarrow$	Clip % $\downarrow$	Color $\times$
A ² CR	16.90	0.81	14.04	8.52	1.92
HSV ² CR	16.61	0.78	14.30	0.57	1.78
C ³ R default	15.08	0.64	15.48	0.00	1.11
C ³ R-v2 frozen	15.84	0.67	14.72	0.00	1.28

5.3 Cylindrical recovery: negative pilot

Table 3 reports the O-HAZE pilot. The validation-selected C³R-v2 disables the noise, disagreement, and hue-guard penalties; these terms therefore receive no empirical support from this dataset. On test, v2 improves over default C³R but A²CR remains better on PSNR in 20/22 images, SSIM in 21/22, and CIEDE2000 in 16/22. C³R yields zero clipped-value pixels and the smallest color expansion, confirming its feasibility/conservatism rather than superior dehazing. This result is why we retain C³R as a complementary branch instead of a separate paper claim.

5.4 CAR scene-08 case study

CAR-Dehaze estimates depth with CAP in HSV but performs the physical inverse in linear RGB. It anchors centered chroma to global airlight, averages chroma at a coarse scale in dense haze, and fuses only low-frequency Lab chroma into a stable luminance backbone. In a GT-defined wall ROI, the hazy input has mean $a^* = -0.445$ and the reference has $a^* = 2.440$. Fixed CAR defaults

recover $a^* = 1.808$, but create visible false-color regions elsewhere. Global reference tuning improves full-image PSNR/SSIM/CIEDE2000 from 17.187/0.633/13.019 to 17.416/0.654/12.188, while weakening the wall to $a^* = -0.031$ (Fig. 2). This demonstrates objective conflict, not a general CAR advantage.

6 Limitations and Reproducibility

The controlled benchmark depends on a synthetic scattering/noise family even though RGB and depth are real. DIODE’s 500 selected frames come from only six physical scene groups. The four paired dehazing datasets were used during development. B9 is roughly 31 times slower than B8 at maximum dimension 96, and the current implementation is CPU-only. B3 isolates a boundary component and is not a complete reproduction of Meng et al. [2]. No claim of state-of-the-art quality or worldwide priority is made. The C³R pilot is low-resolution, non-blind, and lacks a controlled oracle- t/A comparison; its HSV risk is an approximation, not a second physical inversion theorem.

The repository records source URLs, archive hashes, frame manifests, stable seeds, commands, raw CSV files, 10,000-bootstrap outputs, and 50 deterministic unit tests. It also distinguishes raw primary metrics from optional aligned diagnostic metrics. Dataset licenses remain with their authors; generated haze recipes are streamed and not redistributed.

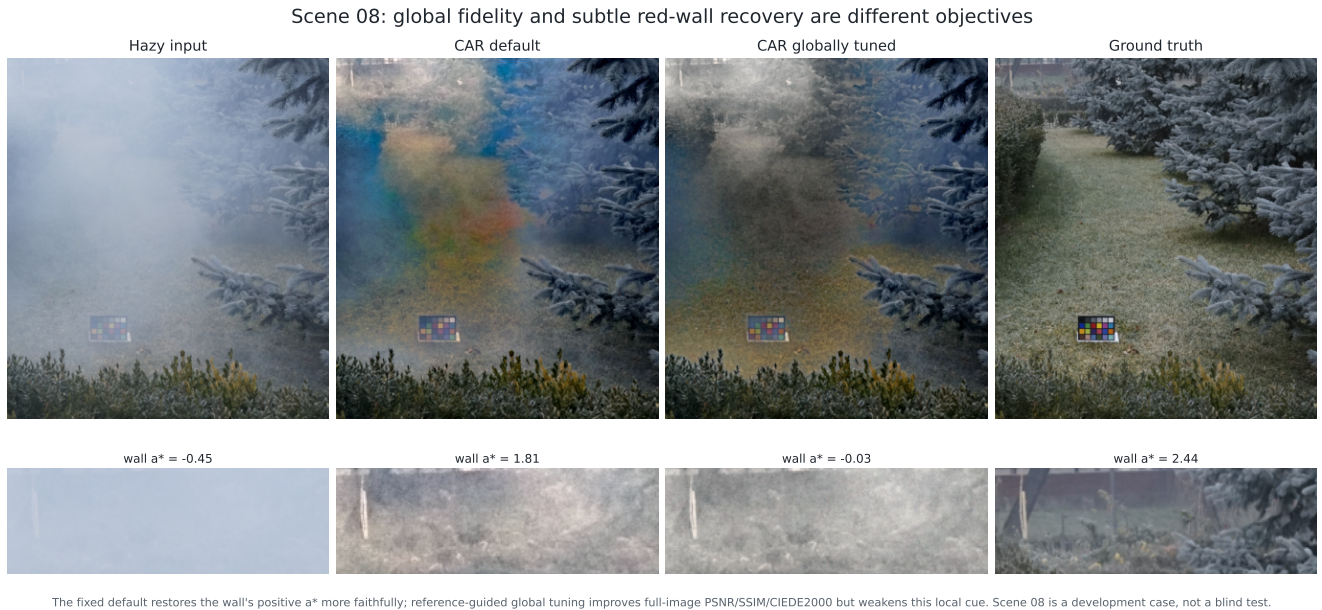


Figure 2: O-HAZE scene 08 development case. The default recovers the red-wall sign but is globally aggressive; reference-guided global tuning improves whole-image metrics but loses the local cue.

7 Conclusion

Airlight-aligned dual gains expose a useful degree of freedom hidden by scalar inverse scattering. Analytic uncertainty risk, an exact RGB-feasible polygon, and joint TV produce strong controlled improvements in structure, noise, and feasibility. The cylindrical companion shows that feasibility can transfer to a circular color representation, but also that a plausible geometry need not improve dehazing: its uncertainty terms hurt the current O-HAZE pilot. Mixed real-paired results, this negative branch, and the CAR case study prevent a universal color or perceptual claim. The next decisive experiment is a pre-registered blind external evaluation, an oracle- t/A RGB-versus-HSV ablation, and comparison with complete external classical baselines.

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